

Motion in One Dimension

→ Direct que

→ Graph based que

$$\text{Avg } \bar{v} = \frac{\Delta x}{\Delta t}$$

$$= \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} v dt$$

$$\text{Avg } \bar{a} = \frac{\Delta v}{\Delta t} = \frac{v_2 - v_1}{t_2 - t_1}$$

$$\Delta x = \int_{t_1}^{t_2} v dt = \text{Area under } v-t \text{ curve.}$$

$$\Delta v = \int_{t_1}^{t_2} a dt = \text{Area under } a-t \text{ curve.}$$

→ Instantaneous vel. $v = f(t)$
 $= f(x) \left] = \frac{dx}{dt}\right.$

use calculus

→ $\underline{\text{Acc}^n}$ → Uniform Acc^n → Speed eqⁿ $\vec{v} = \vec{u} + \vec{a}t$

Inst accⁿ $a = f(t)$
 $= f(x)$
 $= f(v) \left] = \frac{dv}{dt} = v \frac{dv}{dx}\right.$
 use calculus.

$$\vec{S} = \vec{u}t + \frac{1}{2}\vec{a}t^2$$

$$v^2 = u^2 + 2\vec{a}\vec{S}$$

$$\Delta \vec{S}_n = \vec{u} + \vec{a}/2 (2n-1)$$

Physics Galaxy
Videos

Motion in One Dimension

Short-cut tricks for
Cases of uniform ac^n .

These are used when
initial or final vel in
motion is zero.

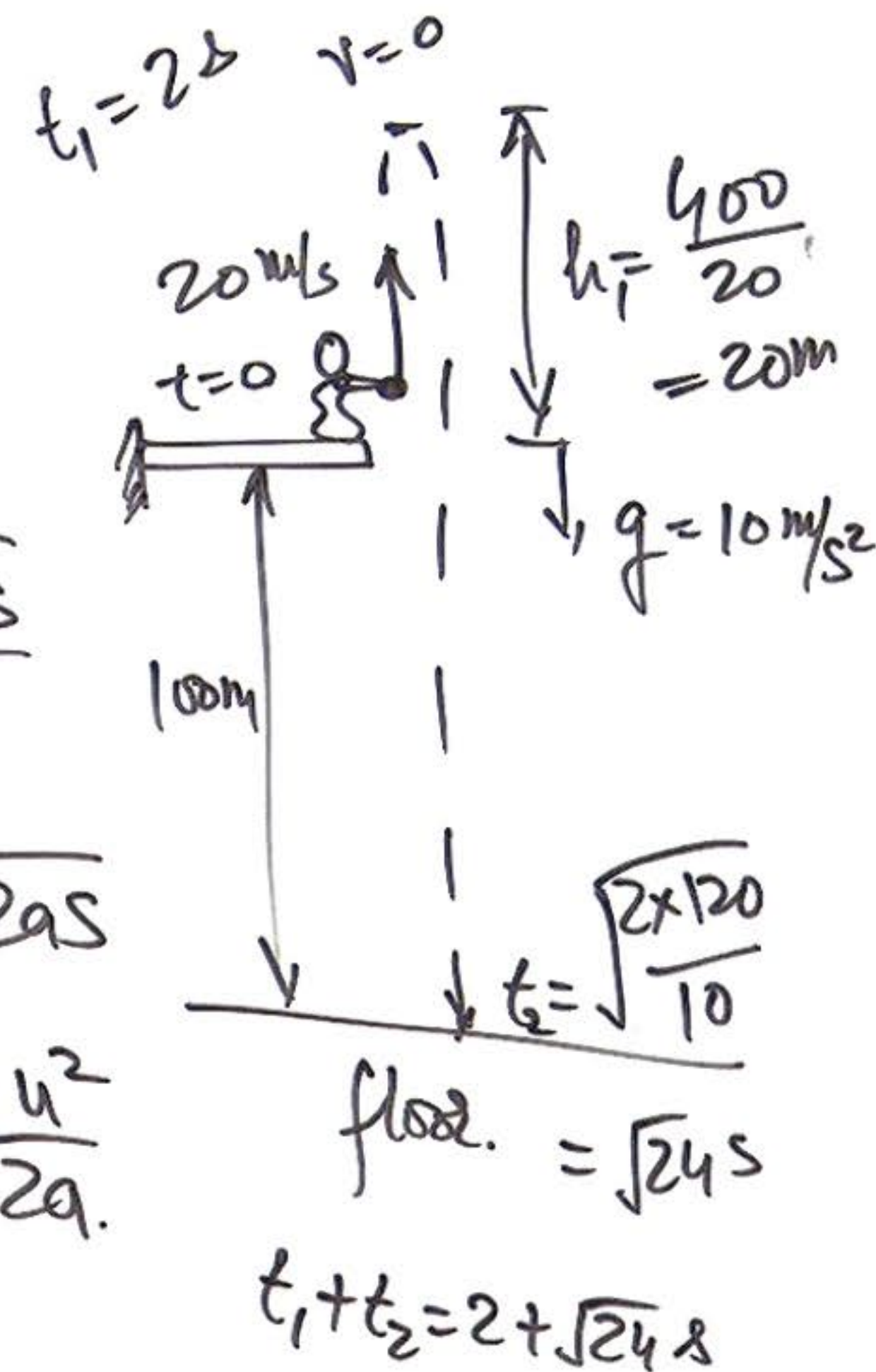
Modified speed ac^n .

$$t = \frac{v}{a} \leftarrow v = at ; u = at$$

$$s = \frac{1}{2}at^2 ; t = \sqrt{\frac{2s}{a}}$$

$$v = \sqrt{2as} ; u = \sqrt{2as}$$

$$s = \frac{v^2}{2a} ; s = \frac{u^2}{2a}$$



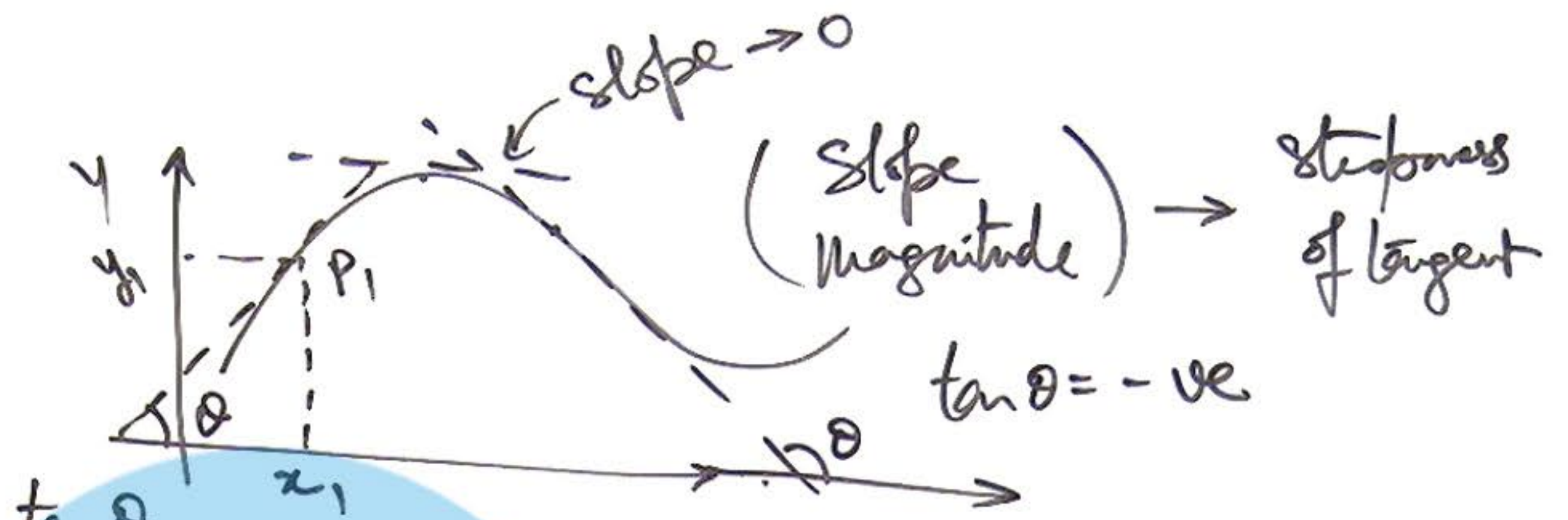
Motion in One Dimension

Graph Based Que-

→ slope → $\frac{dy}{dx}$

→ Area under Curve.

$$Area = \int_{x_1}^{x_2} y \, dx$$



Type of graph

x vs t

v vs t

a vs t

v vs x

a vs x

a vs v

slope

$$\checkmark \frac{dx}{dt} \rightarrow v$$

$$\checkmark \frac{dv}{dt} \rightarrow a$$

$$\frac{da}{dt}$$

$$\checkmark \frac{dv}{dx} \rightarrow \frac{a}{v}$$

$$\frac{da}{dx}$$

$$\frac{da}{dv}$$

